

SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2022

(CBCSS-UG)

Mathematics

MTS 6B 14 (E03)—MATHEMATICAL PROGRAMMING WITH PYTHON AND LATEX

(2019 Admissions)

Time : Two Hours

Maximum : 60 Marks

Section A

*Answer at least eight questions.**Each question carries 3 marks.**All questions can be attended.**Overall Ceiling 24.*

1. List any two data types in python. Also define one variable of each type.
2. Write the python code to print the value of the expression $\frac{ab}{\sqrt{a^2 - b^2}}$ by defining the variables $a = 2^{314}$ and $b = \sqrt[3]{10}$.
3. What is mean by slicing of a string ? Give one example.
4. Define a list of elements $3.14, 0, -2, 1 + \sqrt{2}, 2^{2^{10}}$ and write the code to print the square of third element.
5. Write python code to input the numbers 100, 2.713 to the variables x, e and print their sum and product.
6. Write python code for plotting the points $(0,0), (2,3), (-1, 2)$ on the plane.
7. Write a function to plot a polar curve.
8. Name any two numerical methods for finding the roots of an equation $f(x) = 0$.
9. Write a python program to print the polynomial $2x + 3$.
10. What is mean by documentclass in $\text{\LaTeX}$ Name any two.
11. Write the $\text{\LaTeX}$ command to print the following.

Mathematics, MATHEMATICS

12. Name any two environments in $\text{\LaTeX}$ and explain the purpose.

(8 × 3 = 24 marks)

Turn over

Section B

Answer at least **five** questions.

Each question carries 5 marks.

All questions can be attended.

Overall Ceiling 25.

13. Write a short note on for and while loops in python. Write a python program to print the square root of natural numbers less than 10,000.
14. Write a short note on the matrix creation and operations on matrices using numpy module. Write

a program to print the square of the matrix $\Lambda = \begin{pmatrix} 2 & 3 & 4 \\ 0 & 2.5 & 5 \\ 7 & 6 & 3 \end{pmatrix}$.

15. Write python programs to plot the curves :

(a) $y = x \sin(1/x), 0.001 \leq x \leq 5$ (use 1000 points).

(b) $r = \sin 5\theta, 0 \leq \theta \leq 2\pi$ (use 1000 points).

(c) $x^2 + y^2 = 1, y \geq 0$.

16. Write a short note on the trapezoidal rule to find $\int_a^b f(x) dx$. Write a python program to evaluate

$\int_0^1 \frac{1}{\sqrt{1+x^2}} dx$ using trapezoidal rule with 1000 subintervals of [0,1].

17. Write a short note on Euler's method for solving an initial value problem for first order ordinary differential equations. Write a program to plot the solution curve of the initial value problem

$\frac{dy}{dx} = \frac{x \sin x}{1+x^2}, y(0) = 0$. (use step-size equal to 0.001)

18. Write a $\text{\LaTeX}$ file to produce the following output (without frame).

Derivatives

The derivative of f at c is defined as :

$$f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$

if the limit exists.

Find the derivative of the following functions at the given points :

1. $f(x) = \sin x$ at $c = \pi$.

2. $g(x) = x^3 - x^2$ at $c = \frac{1}{\sqrt{2}}$.

3. $h(x) = e^{-\frac{1}{x}}$ at $c = 2\pi$.

19. Write a $\text{\LaTeX}$ file to produce the following output (without frame).

Hyperbola

The hyperbola with center (h, k) and vertices at $(a, 0)$ and $(b, 0)$ is :

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1.$$

The foci are $(\sqrt{a^2 + b^2}, 0)$ and $(-\sqrt{a^2 + b^2}, 0)$. Sketch the following hyperbolas :

- $9y^2 - x^2 = 36$.

- $16(x+1)^2 - 8(y-3)^2 = 16$.

(5 × 5 = 25 marks)

Turn over

Section C

*Answer any **one** question.
The question carries 11 marks.*

20. (a) Write a short note on conditional statements in python.
(b) Write a python program to input an integer and check whether it is a prime number.
(c) Write a python program to solve the system of equations :

$$\begin{aligned}2x_1 + 3x_2 - x_3 &= 2 \\5x_1 - 2.3x_2 + 8x_3 &= -7 \\10x_1 + 12x_2 - 3x_3 &= 1.\end{aligned}$$

21. (a) Write a short note on 3D Plots using python.
(b) Write a short note on Runge-Kutta method of fourth order.
(c) Write a python program to solve the initial value problem $\frac{dy}{dx} = \cos x, y(0) = 0$ using Runge-Kutta method of fourth order.

(1 × 11 = 11 marks)